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A). **Overview:**

What sets the platform apart :-

For Creators:	For Viewers:
• Complete viewer behavior analysis showing exactly when and why viewers engage or disengage • Advanced pattern recognition revealing which content elements drive the strongest audience response • Predictive analytics forecasting content performance based on historical data patterns • Detailed demographic insights correlated with content preferences • Viewer journey mapping from discovery to long-term engagement • Content impact scoring based on audience retention and growth metrics • Community interaction patterns and engagement hotspots • Comparative analysis across different content formats and styles • Continuous analytics notifications via email/msg	• Personal engagement patterns and content preference insights • Community trend participation metrics • Content discovery patterns • Watch time analysis and content type preferences • Interactive community engagement stats

To-do list : (4th Dec)

- ☑ ~~**Messages**~~
- ☑ ~~**Email & Password Management thru Email**~~
- ☑ ~~**Fun Corner - Analytics & ML**~~
- ☑ ~~**Follower / Following Page for the User**~~
- ☑ ~~**Managing Passwords on Settings Page**~~
- ☑ ~~**Video Playlisting System**~~
- ☑ ~~**Public/Private listing for videos & playlists**~~
- ☑ ~~**Task Queues Implementation**~~
- ☑ ~~**API Development**~~
- ☑ ~~**Caching Strategy**~~
- ☐ **Add Pagination & Filters on FE, BE on Viewer Hist.**
- ☐ **CDN Integration**
- ☐ **Analytics Enhancement**
  - ☐ **Stores followers/following, user-session data**
  - ☐ **Storing & Visualizing videos related metadata**
- ☐ **Continuous Analytics Notification via email/msg**
- ☐ **Unique Vid Recs for the Users wrt vid listings (ML)**
- ☐ **Home Page Caching Strategy**
- ☐ **Deployment Steps**

B). **How the Apps Work Together:**

1.”users” App (User Management) (***Models-User, Profile, Follow, Message***)	
Core Purpose -	Advanced Purpose -
1.Handles user accounts (signup/login) ✔️ 2.Manages user profiles ✔️ 3.Stores user information ✔️ 4.User to User Following feature ✔️ 5.Handles Authentication & Authorization ✔️ 6.Core Authen. & Auth. Security (URL, etc) ❓ 7.User to User Messages feature ✔️ 8.Email & Password Management ✔️ 9.Stores all User App related metadata for Analysis ❓ 10.Continuous analytics notifs via email/msg ❓ ❓ 11.Caching strategy for users related data ✔️	1.Monetization ⁉️⁉️

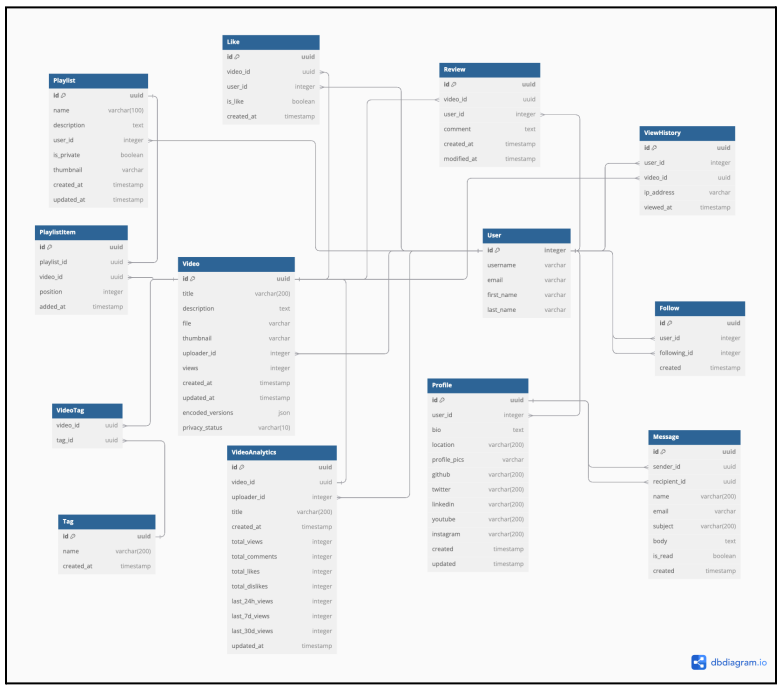
2.”projects” App (Core Features) (***Models-Video, Tag, Like, Review, Playlist, ViewHistory, VideoAnalytics, PlaylistItem)***	
Core Purpose -	Advanced Purpose -
1.Handles video uploads ✔️ 2.Manages video playback ✔️ 3.Stores all relevant video metadata for Analysis ❓ 4.Manages video & playlists listings (public / private) ✔️ 5.Handles video reviews/ratings ✔️ 6.Manages video tagging system ✔️ 7.Fitting the thumbnails with Bars ✔️ 8.Managing user’s viewing history ✔️ 9.Video Compression and Encoding ✔️ 10. Video analytics page ✔️ 11.Task queues for video processing ✔️ 12.Caching strategy for projects Related Data ✔️ 13.CDN integration ❓ ❓ 14.Dynamic Comment Section ✔️ 15.Unique Video Recs for user wrt Vid Listing (ML) ❓ 16.Search Bar and Pagination ✔️ 17.Fun Corner - Analytics and ML ✔️ 18.Handles video playlisting system ✔️	1.Thumbnail Generation ⁉️ 2.Ability to Handle Live Streaming ⁉️⁉️ 3.Security and Compliance (Copyrite) ⁉️⁉️ 4.Creator Tools Page ⁉️⁉️

★ Future Scaling Needs ***(When using multiple servers/instances, high traffic needs, high availability requirements)***	
1.Load balancing ⁉️⁉️ 2.Database Sharding ⁉️⁉️ 3.Optimising Caching Strategy ⁉️⁉️	4.Shifting to Microservices from Monolithic ⁉️⁉️

C). **Transitioning the Code to Industry Standards:**

1. Separate concerns (HTML, CSS, JS)
2. **Clean-up URLs & route them properly** - which pages **need pk to access** user profile, **and/or** which **need "profile = user.request.profile" to access** user profile in views.
3. Make sure only authenticated users are allowed to access particular views in every app.
4. **Clean and debugged** code
5. Adopt modern JavaScript features like fetch, async/await, and template literals.
6. Error handling: **Make sure every interaction has proper success/failure feedback.**
7. Security: **Validate and sanitize inputs.**
8. Responsive design: Use CSS classes to control UI behavior instead of inline styles.
9. **Document everything important** in the codebase
10. Look for the best place to host - AWS free tier, DigitalOcean
11. **When to transfer regular views to API views**:
    A. **If the view**: Only handles data operations (CRUD) ,Doesn't render HTML ,Needs to support multiple clients Gets called by JavaScript/AJAX
    B. **Don't transfer i**f: View renders HTML templates, Handles file uploads with forms, Manages user sessions directly, Does page redirects
12. A. **Cache Data When :** Data is static/semi-static, Operations are expensive, Access frequency is high, Slight staleness is acceptable
    B. **Don't Cache Data When** : User-specific data, Sensitive information, Constantly changing data
    C. **Clear Cache When**: Data is updated, TTL expires, Manual admin clear

D). **DB Schema:** (https://tinyurl.com/dbcshema1)



E). **Key Relationships Between Tables in the DB:**

1) User - Profile Relationship Practical: Each user has exactly one profile, and each profile belongs to exactly one user Technical: One-to-One relationship with Profile.user_id referencing User.id Example: Like a YouTube account (User) having one channel profile (Profile) Access: user.profile or profile.user	2) User - Video Relationship Practical: One user can upload many videos, but each video has one uploader Technical: One-to-Many relationship with Video.uploader_id referencing User.id Example: A YouTuber uploading multiple videos to their channel Access: user.videos.all() or video.uploader	3) User - Review Relationship Practical: One user can write many reviews, each review has one author Technical: One-to-Many relationship with Review.user_id referencing User.id Example: A user commenting on multiple different videos Access: user.reviews.all() or review.user
4) User - Like Relationship Practical: One user can like many videos, each like comes from one user Technical: One-to-Many with Like.user_id referencing User.id Example: A user liking multiple different videos Access: user.video_likes.all() or like.user	5) User - Playlist Relationship Practical: One user can create many playlists, each playlist has one owner Technical: One-to-Many with Playlist.user_id referencing User.id Example: A user creating multiple playlists for different video categories Access: user.playlists.all() or playlist.user	6) Video - Tag Relationship Practical: One video can have many tags, one tag can be on many videos Technical: Many-to-Many relationship using an automatic junction table Example: A video having multiple tags (#funny, #music, #tutorial) Access: video.tags.all() or tag.video_set.all()
7) Playlist - PlaylistItem Relationship Practical: A playlist can contain multiple videos, but a PlaylistItem uniquely identifies a video within the playlist. Technical: One-to-Many with PlaylistItem. playlist_id referencing Playlist.id. Each PlaylistItem corresponds to one playlist. Access: playlist.playlist_items.all() for all PlaylistItems in a playlist.	8) User - Follow Relationship Practical: Users can follow many users and be followed by many users Technical: Many-to-Many through explicit Follow model with additional data Example: Like following multiple YouTube channels Access: user.following_set.all() or user.followers_set.all()	9) Video - Like Relationship Practical: One video can have many likes, each like belongs to one video Technical: One-to-Many with Like.video_id referencing Video.id Example: Multiple users liking or disliking a single video Access: video.likes.all() or like.video

10) Video - Review Relationship Practical: One video can have many comments, each comment belongs to one video Technical: One-to-Many with Review.video_id referencing Video.id Example: Like YouTube comments where multiple users can comment on a single video	11) Video - ViewHistory Relationship Practical: One video can have many view records Technical: One-to-Many with ViewHistory.video_id referencing Video.id Example: Tracking individual views of a video Access: video.views_history.all() Features: Tracks timestamp and IP for anonymous views	12) User - ViewHistory Relationship Practical: Users can have viewing history for multiple videos (nullable) Technical: One-to-Many with ViewHistory.user_id referencing User.id (null=True) Example: Tracking what videos a user has watched Access: user.view_history.all()
13) Video - VideoAnalytics Relationship Practical: Each video has exactly one analytics dashboard tracking engagement metrics Technical: One-to-One relationship with VideoAnalytics.video_id referencing Video.id Example: Like a YouTube Studio analytics panel showing comprehensive video statistics	14) User - VideoAnalytics Relationship Practical: A user can see analytics for all their uploaded videos Technical: One-to-Many with VideoAnalytics.uploader_id referencing User.id Example: A creator viewing analytics across their entire video catalog	15) Message - Profile (Sender) Relationship Practical: One profile can send many messages, each message has one sender (nullable) Technical: One-to-Many relationship with Message.sender_id referencing Profile.id (null=True) Example: Like YouTube's direct messaging where a user can send multiple messages Access: profile.message_set.all() or message.sender
16) Message - Profile (Recipient) Relationship Practical: One profile can receive many messages, each message has one recipient (nullable) Technical: One-to-Many relationship with Message.recipient_id referencing Profile.id (null=True) Example: Like receiving multiple direct messages in your YouTube inbox Access: profile.messages.all() (using related_name='messages') or message.recipient	17) Video - PlaylistItem Relationship Practical: A video can appear in multiple playlists, but a PlaylistItem uniquely identifies the video in a specific playlist. Technical: One-to-Many with PlaylistItem.video_id referencing Video.id.. Access: video.playlist_items.all() for all PlaylistItems referencing a video. playlist_item.video for the video of a PlaylistItem.	

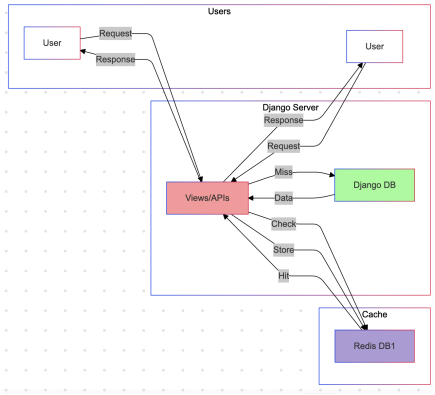
F). **API Endpoints** (with CORS) **:**

API URL	METHODS	What does it do ?	Linked to FE
/api/	GET	Returns list of all available API URLs	❌
/api/users/token/	POST	Get JWT tokens	✅ login form
/api/users/token/refresh/	POST	Refresh JWT token	✅ auth.js
/api/users/token-from-session/	GET	Convert session to JWT	✅ auth.js
/api/projects/	GET	Gets all videos/projects	❌
/api/projects/<id>/	GET	Gets single video/project	❌
/api/debug-videos/	GET	Debug information about all videos	❌
/api/projects/<id>/stream/	GET	Streams video content in chunks	✅ video-player.html
/api/projects/<id>/like/	GET POST	Gets like status and counts, Toggles like/dislike	✅ api_like_dislike.js ✅ api_like_dislike.js
/api/projects/<id>/reviews/	GET POST	Gets all comments for a video Creates new comment	✅ api_comments.js ✅ api_comments.js
/api/projects/<id>/reviews/<review_id>/	PUT DELETE	Updates a comment Deletes a comment	✅ api_comments.js ✅ api_comments.js
/api/projects/<id>/reviews/stream/	GET	Streams comments (real-time)	✅ api_comments.js
/api/users/<id>/follow/	POST	Toggle follow/unfollow & return stats	✅ api_follow.js ✅ api_profile_follow.js

G). **Video Analytics and User Behavior Patterns:**

i. Views Based for Creators	ii. Follower Based for Creators	iii. User Behaviour Pattern
A. Total Views of all vids ✔️ B. Daily Views for each vid ✔️ C. Hourly Views for each vid ✔️	A. Follower growth rate ❓ B. Follower engagement metrics ❓ C. Follower retention statistics ❓	A. User session data ❓ B. Navigation patterns ❓ C. Search behavior ❓ D.Time spent on platform ❓
iv. Engagement Based for Creators	**v. Content Performance Metrics for Creators**	
A. Total like-dislike for all videos ✔️ B.Total Comments for all videos ✔️ C. Total shares for all videos ❓ D. Watch time patterns ❓ E. Viewer retention rates ❓ F. Engagement/drop-off points ❓	A. Video completion rates ❓ B. Average watch duration ❓ C. Peak viewing times ❓ D.Audience retention graphs ❓	

H). **Caching Strategy:**



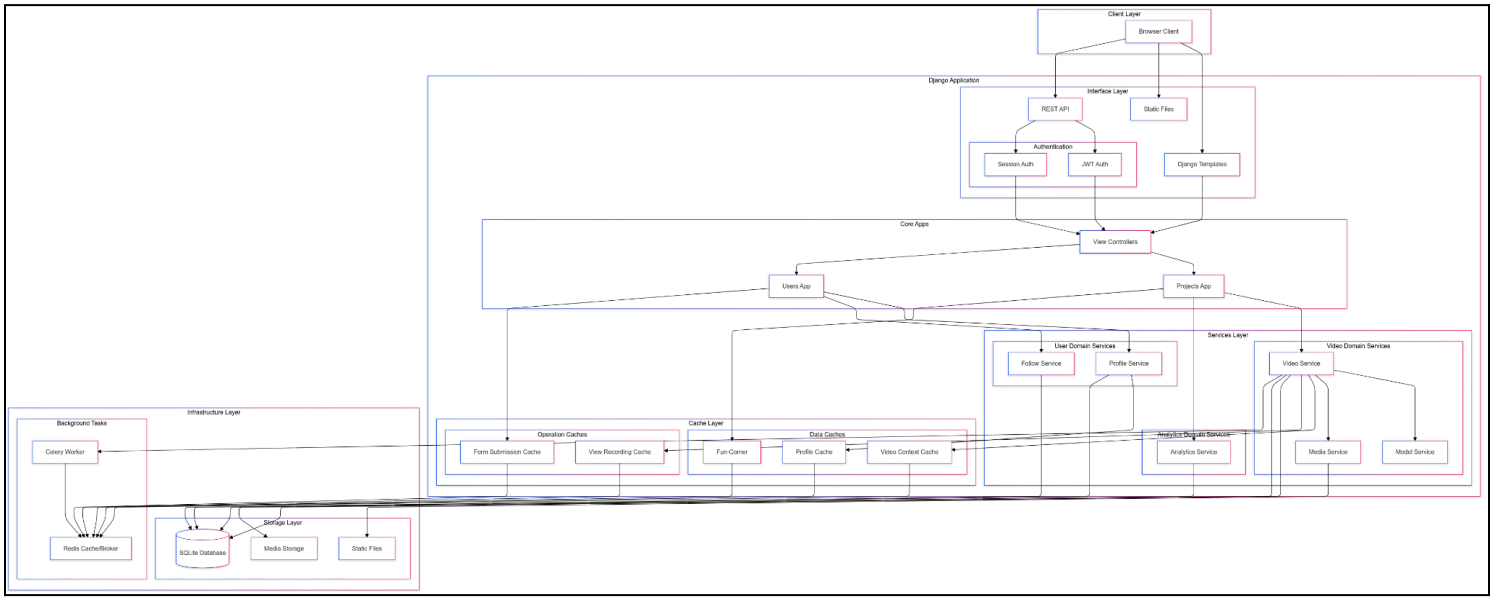
Server Side Global Redis Caching

1. Multiple users hitting Django server
2. Server checks **Redis Global Cache** first (DB1)
3. Cache hit → return data
4. Cache miss → get from Django DB
5. Store fresh data in cache
6. Return to user

Rules for the Uncached Views in Users App				
🟢 Cache Duration: (Data Type and Duration)	🟢 Monitor: (Cache & DB)	🟡 Cache only after Monitoring (future)	🔴 Don't Cache: (Sensitive Data, Too User-Specific)	
Real-Time Data: 1-5 minutes Regular Data: 1 hour - 10 mins Static Data: 24 hours or NONE	Cache hits/misses Cache Memory usage Cache Error Rates DB Queries DB “Read/Write” ratio	inbox view:	login, logout_user Register_user clear_temp_tokens user_videos viewer history	Update_profile , Create_message, View_message Creator_studio: analytics:, settings:

Type of Data being Cached in Redis (DB1)	**Cache Function**	**Policy**	**TTL**
Major Studio Box Office Data (/fun_corner)(Static)	get_studio_data	Cache-Aside	24 hrs
Current Movies Playing Data (/fun_corner)(Static)	get_current_movies	Cache-Aside	24 hrs
Movie Recs Trained Model (/fun_corner)(Static)	get_movie_recs_model	Cache-Aside	24 hrs
Tracking & Recording Views of vid (/project)(Real-Time)	can_record_view	Time-Based	5 mins
Context of a Vid (title,tags,desc.,etc)(/project)(Regular)	set_context	• Read-Through • Cache Invalidation on Write	15 mins
Form key to stop multiple submit (All form submit views)(Real-Time)	set_submission	Time-Based	1 min
Profile Info of User(name,socials,etc,)(/profile)(Regular)	set_profile_cache	• Read-Through • Cache Invalidation on Write	10 mins

I). **System Architecture:** (https://tinyurl.com/sys-archew)



This architecture represents a classic monolithic design where all components are tightly integrated within a single Django application. Here's why it's monolithic:

1. Single Application Integration: All components (Web Layer, Core Apps, Services) exist within one Django project, Services directly communicate with each other within the same application space, Shared database and storage resources, Single deployment unit.

2. Vertical Structure: The architecture flows from top (Client) to bottom (Storage), Each layer directly depends on the layers below it, Tightly coupled components where Services can directly access the database

3. Shared Resources: Single database (SQLite) serving all services, Shared media storage for all file operations , Common Redis instance for caching and message brokering, Unified authentication system (JWT and Session)

SWITCH FROM MONOLITH TO MICROSERVICES ONLY WHEN THE TEAM GROWS IN SIZE AND DEVELOPERS START STEPPING ON EACH OTHER'S TOES. IT'S FINE TO START AS A MONOLITH AND THEN SWITCH EVENTUALLY.

This monolithic approach has advantages for this application:

- Simpler development and deployment, Easier debugging as all code is in one place
- Direct database access without network overhead
- Simpler transaction management
- Suitable for the application size and complexity
- The architecture is well-organized within its monolithic constraints, using clear separation of concerns (Web, Services, Storage) while maintaining the benefits of a unified system.

J). **ENV Variables:**

SMTP	
1) Google SMTP EMAIL_HOST = 'smtp.gmail.com' EMAIL_PORT = 587 EMAIL_USE_TLS = True EMAIL_HOST_USER = 'filmsnation99@gmail.com' EMAIL_HOST_PASSWORD = 'iipo pjsw mhjv aumx'	2) Mailtrap SMTP EMAIL_HOST = 'sandbox.smtp.mailtrap.io' EMAIL_PORT = 2525 EMAIL_USE_TLS = True EMAIL_HOST_USER = '325d9889579e2a' EMAIL_HOST_PASSWORD = '51ca0096774d01'

K). **GitHub WorkFlow:**

<table>
<tr>
<td rowspan="2">+ To get the exact repository state from remote to your local

Check current branches (*indicates current branch)
> git branch

First switch to remote branch (if not already there)
> git checkout branch-name

Fetch all changes from remote
> git fetch origin

Hard reset to the remote branch
> git reset --hard origin/branch-name

(if needed)

Remove the database
> rm db.sqlite3

Remove all migration files except __init__.py
> rm app-name/migrations/0*.py

Make new migrations and apply them
> python manage.py makemigrations
> python manage.py migrate

Create a superuser
> python manage.py createsuperuser</td>
<td>1.Initial Status and Feature Branch Work

Check current branches (*indicates current branch)
> git branch

Stage all changes
> git add .

Commit changes with message
> git commit -m "your message"

Push feature branch to remote
> git push origin feature/branch-name</td>
<td>2. Update Main Branch

Switch to main branch
> git checkout main

Pull changes from develop to main
> git pull origin develop

Push changes to main
> git push origin main

#If rejected:
sync with remote main
> git pull origin main

try pushing again
> git push origin main</td>
<td>3.Update Develop Branch

Switch to develop
> git checkout develop

Get latest changes
> git pull origin develop

Merge feature branch into develop
> git merge feature/branch-name

Push updated develop
> git push origin develop</td>
</tr>
<tr>
<td>4.Clean Up Feature Branch

Delete local feature branch
> git branch -d feature/branch-name

Delete remote feature branch
> git push origin --delete feature/branch-name</td>
<td>5.Start New Feature Branch

Create and switch to new feature branch from develop
> git checkout develop
> git checkout -b feature/new-branch-name

Push new branch to remote and set tracking
> git push -u origin feature/new-branch-name</td>
<td>6.Final Sync

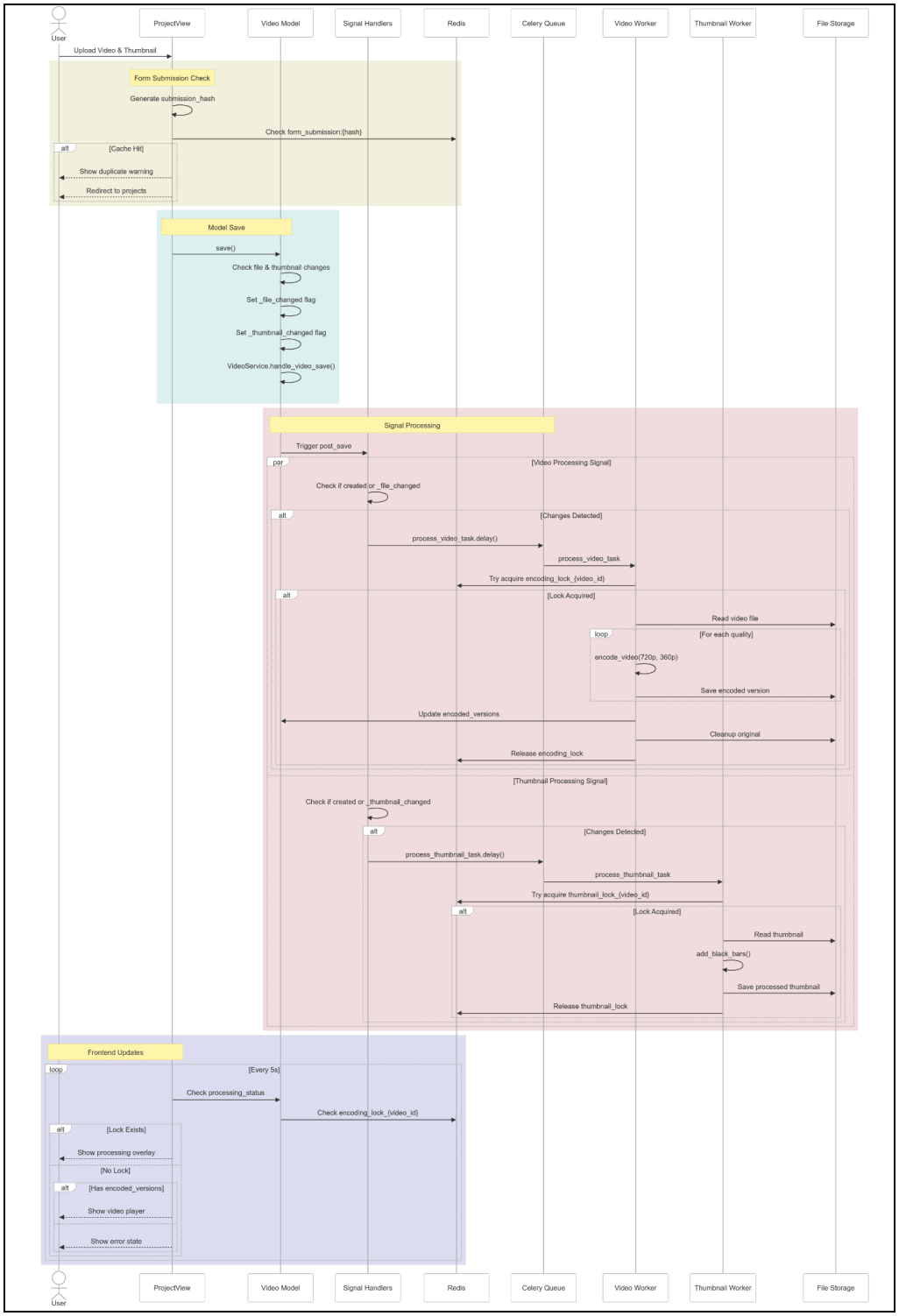
Get all remote updates
> git fetch --all

Pull updates for all branches
> git pull --all

Verify branch structure
> git branch</td>
</tr>
</table>

L). **Functionalities Explanation:**

- **Video Upload Pipeline with task queue Implementation**
  (https://tinyurl.com/videoupload122)



- **Authentication using JWT & Session cookies**

The authentication system uses a hybrid approach combining both traditional Django sessions and JWT tokens. Here's how it works:

When a user logs in through the login_user view, two things happen simultaneously. First, Django's session authentication is initiated with login(request, user), which creates a session cookie for the user. At the same time, JWT tokens (both access and refresh) are generated using RefreshToken.for_user(user) and sent back to the client in a JSON response. These tokens are then stored in the browser's localStorage.

For ongoing authentication, your system is flexible. Traditional Django views can rely on the session cookie for authentication, while API endpoints can use the JWT tokens. This is why your API views have both JWTAuthentication and SessionAuthentication in their authentication classes. When making API requests, the frontend includes the JWT token in the Authorization header, but the session cookie is also present for traditional requests.
During logout, handled by logout_user, the system cleans up both authentication methods. The Django session is cleared using logout(request), and a special cookie is set that triggers JavaScript to clear the JWT tokens from localStorage. This ensures both authentication methods are properly terminated.

The registration process implements a hybrid authentication approach. When a user registers through register_user, it generates both a traditional Django session with login(request, user) and JWT tokens using RefreshToken.for_user(user). The tokens are temporarily stored in the session and then transferred to localStorage via a script in the profile update template. This ensures the user has immediate access to both regular web views (via session) and API features like likes, comments, and follows (via JWT) without needing to log in separately. The system seamlessly handles both web and API authentication from the moment of registration.

- **Video Streaming Implementation Process**

Our video streaming implementation follows a sophisticated process that efficiently delivers video content from server storage to end users. The process begins when a user requests a video through our web interface. Instead of loading the entire video file into memory, as was done in our previous implementation, we now utilize a chunked streaming approach.
When the video player initializes, it first requests only the video metadata. This metadata includes information about the video duration, dimensions, and available quality versions. The server, through our VideoService, determines the appropriate version of the video to serve, typically defaulting to 720p if available, with fallback options to other qualities or the original file if necessary.

As playback begins, the video player sends HTTP range requests to our server. These requests specify exactly which portions of the video file are needed at any given moment. Our stream_video view function processes these requests, using the file_iterator to read the video file in small, manageable chunks of 8KB (8192 bytes). This chunked approach means that only a small portion of the video is held in memory at any given time, significantly reducing server resource usage.

The file_iterator function is central to our streaming process. When it receives a range request, it opens the video file, seeks to the specified position, and yields data in controlled chunks. This streaming process continues throughout the video playback, with the player requesting new chunks as needed. For seeking operations, when a user jumps to a different part of the video, the player simply sends a new range request for that specific portion of the file.

Our implementation includes built-in support for quality management through encoded versions of each video. The ModelService handles the URL generation for these different quality versions, while the VideoService manages the overall video delivery process, including view counting and caching of video context data. This service-oriented architecture ensures clean separation of concerns and maintainable code.

The actual delivery of video content to the client is handled through Django's StreamingHttpResponse, which maintains an open connection to stream the video chunks. The response includes proper HTTP headers for content range and length, allowing the client's video player to properly handle the incoming stream. This streaming response is crucial for efficient memory usage and proper handling of large video files.

On the frontend, we utilize the Video.js player, which is configured to handle these chunked streams efficiently. The player maintains its own buffer of video data, requesting new chunks as needed based on the user's playback position and network conditions. Our custom CSS ensures proper video display regardless of the video's original dimensions, maintaining aspect ratios while providing a consistent user interface.

This entire process represents a significant improvement over traditional video serving methods. Where previously we would load entire video files into memory, potentially causing server strain with multiple concurrent users, we now handle video delivery in a memory-efficient, scalable manner. The chunked streaming approach, combined with quality management and proper range request handling, provides a smooth, efficient video playback experience while maintaining server stability under load.

The end result is a robust video streaming system that efficiently manages server resources while providing users with smooth, high-quality video playback. The system handles everything from initial video request through continuous playback and seeking operations, all while maintaining minimal memory footprint and optimal performance characteristics.

● Dynamic Comment Section

The dynamic comments system represents a sophisticated approach to real-time user interaction in our video platform. Unlike traditional comment systems that require page refreshes, our implementation provides a seamless, single-page application experience while maintaining data consistency and user engagement.

When a user first loads a video page, the system initializes by embedding existing comments directly into the page's HTML. This initial state is cleverly stored in a hidden input element as JSON data, allowing immediate access to comments without additional server requests. This approach significantly reduces initial load time and provides a smooth user experience from the moment the page loads.

The heart of our commenting system lies in its event handling architecture. Rather than attaching individual event listeners to each button or element, we implement a central event delegation system. This system captures all click events at the document level and determines the appropriate action based on the clicked element. This approach not only improves performance but also handles dynamically added content without requiring additional event binding.

Comment creation follows a sophisticated process. When a user clicks the "Add Comment" button, the system smoothly reveals a comment form. As the user types and submits their comment, our JavaScript handler intercepts the form submission, preventing the default page reload. Instead, it packages the form data and sends it to the server via an AJAX request. The server processes this request, stores the comment in the database, and returns the formatted comment data. The frontend then seamlessly inserts this new comment into the page's DOM, providing immediate visual feedback to the user.

The editing and deletion processes follow similarly sophisticated patterns. For comment owners, edit and delete buttons appear alongside their comments. Clicking edit transforms the comment text into an editable form, while delete triggers a confirmation modal. Both actions communicate with the server asynchronously, updating the UI only after successful server responses. This ensures data consistency while maintaining a responsive user interface.

Our backend architecture centers around the Review model, which stores essential comment data including the comment text, user reference, video reference, and timestamps. Each comment is assigned a unique UUID, ensuring reliable identification across the system. The model's structure supports efficient querying and maintains proper relationships between users, videos, and comments.

The server-side implementation includes robust view functions that handle comment creation, updates, and deletion. These views incorporate proper authentication checks, ensuring that users can only modify their own comments. They return JSON responses that the frontend can easily process and display. The system also includes protection against common issues like double submissions and race conditions.

The frontend JavaScript code is organized around clear responsibilities. The createCommentRow function handles the HTML generation for comments, ensuring consistent formatting. The handleSubmit function manages form submissions, including preventing double submissions and providing user feedback. The system also includes sophisticated error handling, displaying appropriate messages to users when issues occur.

This entire system works together to provide a modern, responsive commenting experience. Users can interact with comments instantly, seeing their changes reflected immediately while the system ensures all data is properly synchronized with the server. The implementation balances performance, user experience, and data integrity, creating a robust platform for user engagement through comments.

The result is a commenting system that feels natural and responsive, encouraging user interaction while maintaining data consistency and system reliability. Through careful architecture and modern web technologies, we've created a seamless experience that enhances the overall video platform functionality.

- **User Generated Data Visualization Pipeline**

When a user requests to view analytics, the flow begins at the analytics view function in views.py, which first validates user permissions by comparing the requested profile with the current user's profile. Upon authorization, it triggers ProfileService's get_analytics_data method.

The ProfileService initiates data gathering by retrieving the user's profile and their uploaded videos, then delegates to VideoService's get_creator_studio_analytics method, which coordinates the creation of five distinct charts: total views, daily views, hourly distribution, likes/dislikes, and comments.

VideoService uses AnalyticsService in two ways:

1. For aggregate metrics (total views, likes/dislikes, and comments), it pulls from the denormalized VideoAnalytics table, providing efficient access to pre-calculated totals.

2. For time-based analytics (daily views and hourly distribution), it queries the views_history relationship table, as these require detailed timestamp data.

The data flows back through the service layers: AnalyticsService formats the raw data, VideoService combines it into structured chart datasets, ProfileService packages it, and the view passes it to the template. The template then converts this data to JSON for Chart.js to render the visualizations.

This hybrid approach - using denormalized data for totals and relationship queries for time-series data - optimizes performance while maintaining data accuracy. The service layer architecture ensures clean separation of concerns and maintainable code.